
EXERCISES

Exercise sheet 7

AI504 — Knowledge Representation

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1 Exercise 1⁺

Write the update rule for stochastic gradient with mini-batches of size m on a generical machine learning model $y = h(x)$ and with loss function L . Write the update formula with momentum in the case of mini-batch of size m .

1.1 Solution

stochastic gradient with mini-batches of size m with minimization of the loss function $L(h(x), y)$

$$w_{k+1} = w_k - \frac{\alpha_k}{m} \sum_{i \in B_k} \nabla_w L(h(x_i; w_k), y_i)$$

with momentum

$$w_{k+1} = w_k + v_{k+1}$$

where

$$v_{k+1} = \beta v_k - \frac{\alpha_k}{m} \sum_{i \in B_k} \nabla_w L(h(x_i; w_k), y_i)$$

2 Exercise 2^{*}

In a regression task we assume $h(x; w) = w_0 + w_1 x_1 + w_2 x_2 + \dots + w_d x_d$. For the estimation of the parameters $w \in \mathbb{R}^{d+1}$ using the examples $\{(x_1, y_1), \dots, (x_n, y_n)\}$ we can use the least squares loss function

$$\min R_n(w) = \sum_{i=1}^n L(h(x_i; w), y_i) = \min \|y - Xw\|_2^2$$

Where

$$X = \begin{bmatrix} 1 & x_{11} & x_{21} & \dots & x_{d1} \\ 1 & x_{11} & x_{21} & \dots & x_{d1} \\ \vdots & \ddots & & & \\ 1 & x_{1n} & x_{2n} & \dots & x_{dn} \end{bmatrix}$$

This problem admits a closed form solution by means of the normal equations $w = (X^T X)^{-1} X^T y$. You find the derivation of this result in these slides from DM579/AI511. The L_2 Regularized risk is

$$\min_w R_n(w) + \lambda \|w\|_2^2 = \sum_{i=1}^n L(h(x_i; w), y_i) + \lambda \sum_{j=0}^d w_j^2 = \|y - Xw\|_2^2 + \lambda \|w\|_2^2$$

admits also a closed-form solution: $w = (X^T X + \lambda I)^{-1} X^T y$

Provide a computational analysis of the cost of computing the estimates of w by means of these closed-form solutions and compare these costs with the cost of carrying out the gradient descent. When is the gradient descent faster?